

CERN celebrates 70 years of scientific discovery and innovation

CERN will turn 70 on 29 September 2024. A rich celebration programme will enable the public to celebrate CERN's illustrious past and engage with its bright future



On 25 January, CERN, the European Laboratory for Particle Physics, announced a programme to celebrate its 70th anniversary in 2024. This landmark year honours CERN's remarkable contributions to scientific knowledge, technological innovation and international collaboration in the field of particle physics. Throughout the year, a variety of events and activities will showcase the Laboratory's rich past as well as its bright future.

Leading up to an official high-level ceremony on 1 October, the preliminary anniversary programme, spanning the entire year, offers a rich array of events and activities, aimed at all types of audiences, at CERN and in the Organization's Member States and Associate Member States and beyond. The first public event, scheduled for 30 January, will combine science, art and culture, and will feature a panel of eminent scientists discussing the evolution of particle physics and CERN's significant contributions in advancing this field. On 7 March and 18 April, special events will showcase the practical applications of high-energy physics research in everyday life. Mid-May will see a focus on the importance of global collaboration in scientific endeavours, while the events in June and July will explore the current unanswered questions in particle physics and the facilities being planned for future breakthroughs. From talks by distinguished scientists and exhibitions showing CERN's cutting-edge research and the diversity of its science and its people, to public engagement initiatives worldwide, everyone will find something to enjoy in this programme.

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“CERN’s achievements over the 70 years of its history show what humanity can do when we put aside our differences and focus on the common good”, says Fabiola Gianotti, CERN Director-General. “Through the celebrations of CERN’s 70th anniversary, we will demonstrate how, over the past seven decades, CERN has been at the forefront of scientific knowledge and technological innovation, a model for training and education, collaboration and open science, and an inspiration for citizens around the world. This anniversary is also a great opportunity to look forward: CERN’s beautiful journey of exploration into the fundamental laws of nature and the constituents of matter is set to continue into the future with new, more powerful instruments and technologies.”

CERN came to life in 1954, in the aftermath of the Second World War, to bring excellence in scientific research back to Europe and to foster peaceful collaboration in fundamental research. This collective effort has pushed back the frontiers of human knowledge and of technology. As more powerful accelerators and experiments were built, foundational discoveries and innovations were made: among others, the multiwire proportional chamber was invented in 1968, the neutral currents were discovered in the 1970s, the W and Z bosons were discovered in 1983, the precision measurement of the Z boson and of other parameters of the electroweak theory was made in the 1990s thanks to the Large Electron Positron (LEP) collider, the Large Hadron Collider started up in 2009, and the Higgs boson was discovered in 2012. CERN is also the birthplace of the World Wide Web and has generated technologies that

are used in other fields, including medical diagnostics and therapy and environmental protection.

Today, CERN counts 23 Member States, 10 Associate Member States and a vibrant community of 17,000 people from all over the world, with more than 110 nationalities represented. Currently, the Laboratory is home to the Large Hadron Collider, the world’s most powerful particle accelerator. Building on its remarkable legacy of research and technological development, CERN is already looking to the future, in particular by studying the feasibility of a Future Circular Collider.

“This anniversary year is for everyone and should engage and inspire scientists, policy makers and the public. We are looking forward to welcoming everyone at CERN for the many events being planned, but also to the celebrations in our Member States, Associate Member States and beyond”, says Luciano Musa, coordinator of the CERN 70th anniversary. “These international events are a testament to CERN’s impact on scientific knowledge, technological development and worldwide collaboration.”

CERN extends an invitation to everyone to take part in these inspiring events, which aim to kindle scientific curiosity, honour scientific progress and collaborative efforts, and underscore the role of science in society. Join us in this year of celebration as we honour our glorious past and shape a bright future for CERN and its community. For the complete CERN70 anniversary events and programme of activities, please visit: cern.ch/cern70.

For more information, read our [media kit](#).

CERN70: The Laboratory takes shape

Part 2 of the [CERN70](#) feature series : Franco Bonaudi, one of the pioneers of CERN's accelerators, looks back at the Laboratory's early years, during which everything had yet to be invented



On 17 May 1954, the ground-breaking ceremony for the construction of the European Laboratory for Particle Physics took place on the Meyrin site in Geneva, before the eyes of Geneva civil servants and CERN staff. (Image: CERN)

Geneva was chosen as the site of the CERN Laboratory at the third session of the Provisional Council in 1952. The city's central position in Europe, its neutrality during the war and the fact that it was already home to international organisations played a decisive role.

Construction of the Laboratory began in 1954. While waiting for their premises, the first members of the CERN community moved into the Geneva Institute of Physics and worked from a building and barracks near the airport. The theory group had already been founded in 1952 at the Institute of Theoretical Physics in Copenhagen alongside the Nobel Prize-winning physicist Niels Bohr. It moved to Geneva in 1957.

Two accelerator projects were launched in parallel: an innovative accelerator of unprecedented power, the Proton Synchrotron (PS), and, pending its commissioning, a more conventional machine, the Synchrocyclotron (SC). The SC, built in just three years, went into service in 1957, enabling physicists to get CERN's first experiments up and running quickly.

Recollections

Franco Bonaudi became involved in the great CERN adventure before the Organization was even founded. Given the responsibility of coordinating the first accelerator project, the Synchrocyclotron,

he supervised the construction work. He went on to take part in the construction and operation of many accelerators at CERN and held the post of Director of Accelerators from 1976 to 1978.

"In 1951, I was at the Turin Polytechnic when Edoardo Amaldi, one of CERN's founding fathers, contacted the teaching staff to find out whether they knew of any young scientists who would be interested in taking part in the construction of a particle physics laboratory. I went to Rome to meet Giuseppe Fidecaro, Amaldi's right-hand man, and it was there that I learnt that the idea was to build a synchrocyclotron. I agreed to participate and left for Liverpool, where I took part in the design studies.



The Synchrocyclotron building in November 1955, during its construction. (Image: CERN)

We worked in various European institutes for two years without even knowing where the machine was going to be built. We all started together in huts at Geneva airport in 1954. At that time, the Laboratory consisted of about 150 people of all nationalities. We quickly got used to working together and communicated with each other in bad English. We felt rather lonely, being new to the area, and so we became very close. I

remember that the CERN phone directory contained everyone's home number. Some real friendships were formed.

My colleague Joop Vermeulen and I were the first to move onto the site of the new Laboratory. We erected a hut with windows, right in the middle of the worksite, and stayed there for two years. The cyclotron was completed in only three years, which was fantastic. We had great enthusiasm and were often required to be inventive. I remember, for example, that we had no heating during our first winter at the SC. We managed to borrow some cast-iron resistors from the Geneva trams. Thanks to this improvised electric heating system, we were able to keep warm all winter. I could tell you dozens of stories like that ..."



Transport of a Synchrocyclotron coil through the village of Meyrin in May 1956. (Image: CERN)

This interview is adapted from the 2004 book "Infinitely CERN", published to celebrate CERN's 50th anniversary. Franco Bonaudi [passed away](#) in 2008 at the age of 80, read more about him in the [CERN Courier](#).

BioDynaMo: cutting-edge software helps battle cancer

A novel mathematical model developed as part of the BioDynaMo project, born from CERN openlab, mimics vascular tumour growth in breast cancer and its response to treatment

Choosing the right cancer treatment is a massive undertaking involving multiple stages, high experimental complexity and significant costs. Currently, two main methods are used to find the best possible treatment solutions: in vitro testing

and clinical trials. However, predicting the drug effects on each individual patient remains the Holy Grail of personalised medicine.

Born from CERN openlab in the CERN IT department, BioDynaMo is an innovative tool for

“in silico” testing, i.e. experimentation carried out on a computer. Based on mathematical models, it creates and runs complex 3D computer simulations that help understand cancer progression and identify the most effective treatment strategies for specific tumour cases.

In a recent scientific publication, scientists affiliated with CERN, the Technical University of Munich and the University of Texas at Austin demonstrated the significant potential of advancing medical therapy with the help of BioDynaMo. The model successfully replicates medical data on recorded tumour growth and the effects of two anti-cancer drugs, Doxorubicin and Trastuzumab. By fitting the BioDynaMo models to the available pre-clinical data, scientists proved the platform’s ability to simulate different levels of efficacy of various drugs, treatment combinations and dosage regimens.

BioDynaMo is an open source project that strives to provide the most efficient and performant

simulation platform for agent-based models. It accommodates a diverse range of use cases and can address research questions in oncology, neuroscience, epidemiology and many more disciplines. With its ability to simulate almost two billion agents (or cells), BioDynaMo is a powerful tool for analysing many different complex systems. Since 2015, BioDynaMo's consortium of scientists has been working on developing and optimising the engine, improving its performance and usability. For more information, click [here](#).

BioDynaMo's inception was funded with the support of CERN’s budget for knowledge transfer for the benefit of medical applications and the project continues to be supported by the CERN and Society Foundation. Find

out how you can support the BioDynaMo project here: <https://www.biodynamo.org/contacts-us/support-biodynamo>.

Marina Banjac

HiLumi News: protecting the components of CERN’s future accelerator

The collimation system of the Large Hadron Collider (LHC), which protects the accelerator’s components, needs an upgrade to be able to handle the performance of CERN’s future accelerator

A major upgrade of the collimation system of the Large Hadron Collider (LHC) began during the first long shutdown of CERN’s accelerator complex (LS1, 2013–2015) and continued during LS2 (2019–2021), in preparation for the High-Luminosity LHC (HL-LHC). As its name suggests, the HL-LHC will surpass the LHC in terms of luminosity, i.e. the number of collisions that take place within the LHC experiments. The accelerator’s equipment therefore requires enhanced protection, which is where the collimation system comes in.

What is a collimator?

Collimators are movable blocks made of materials that can absorb particles. Shaped like jaws, they close tightly around the beam to clean up particles that stray from their path. The materials used for these jaws and their various components are capable of withstanding extremes of pressure and temperature, as well as high levels of radiation.



These two new collimators have been developed at CERN for the future HL-LHC. These models will be installed at LHC interaction points 1 (ATLAS detector) and 5 (CMS detector) during Long Shutdown 3 (LS3). (Image: CERN)

Why do beams need cleaning?

Particles that stray from the beam path could collide with sensitive accelerator components, such as superconducting magnets, and interfere with their operation or, in the worst case, damage them. To prevent this from happening, collimators are placed at strategic locations around the LHC ring, where they either absorb stray particles or deflect them towards beam dumps. Protection is particularly crucial in the vicinity of the experiments, where the beam size is reduced to increase the chances of collision.

The LHC currently has 118 collimators of different kinds. The future HL-LHC will have 126 collimators, including brand new models custom made at CERN. Recently, two new prototypes (TCLPX and TCTPXH) have been successfully developed and tested, under the supervision of François-Xavier Nuiry, engineer in charge of the HL-LHC collimator production. Destined for LHC interaction points 1 (ATLAS detector) and 5 (CMS detector), they are double-beam collimators. This optimised configuration enables two beams (circulating in opposite directions) to pass through the same

vacuum chamber, thus freeing up space for the collimators' jaws, which are thicker and more powerful in this location. "These two prototypes are innovative in several ways," explains Dylan Baillard, a mechanical engineer in CERN's Targets, Collimators and Dumps section. "They are fitted with a remote alignment and levelling system, which helps reducing the radiation dose received by the teams working on them. The collimator flanges can be connected and disconnected more easily thanks to integrated connection tools. Finally, ion pumps are used to ensure an excellent vacuum quality because the collimators, which are close to the beams, always operate in a vacuum and must not disrupt the circulation of the beams."

The final tests were successfully completed in December, and series production of the two new types of collimator should begin this year. Twelve double-beam collimators will be installed in the machine during Long Shutdown 3 (LS3, 2026–2028).

Anais Schaeffer

Accessible by design: how CERN Science Gateway exhibitions are tailored to people with visual impairments

Visitors from the Geneva association for visually impaired people (ABA) test the accessibility of the new exhibitions



A visitor from ABA testing the Discover CERN: Accelerate exhibition with her guide. (Image: CERN)

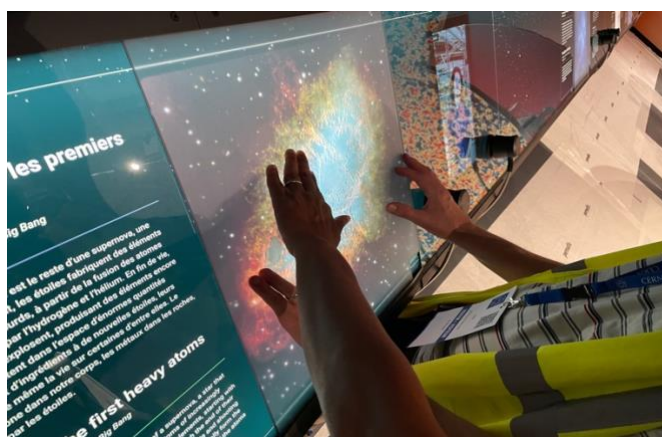
Since opening in October, CERN Science Gateway has welcomed almost 100 000 visitors. Among

these was a group of around 50 people from the [Association pour le Bien des Aveugles et malvoyants](#) (ABA), the reference association in Geneva for blind and visually impaired people, who toured the exhibitions on 28 November. ABA has worked with the CERN exhibitions team since 2019 and this was the chance for its members, including some visually impaired members, to see the fruit of this collaboration.

At the event, Emma Sanders, head of the CERN exhibitions team, and Bernard Jost, ABA's accessibility project manager, explained how they had developed the exhibitions with accessibility and inclusivity at the forefront.

"We learnt very early on that making content more accessible usually benefits more than one

kind of audience group,” Emma explained. “Tactile content is great for people who are blind or who have visual impairments, but it is also appreciated by many children. Wheel-chair accessible furniture is also good for parents pushing kids in buggies.” “The idea came from CERN,” says Jost. “While the law requires cultural sites in Geneva to be as accessible as possible, what’s praiseworthy about CERN Science Gateway is that they made a point of taking accessibility into account from the beginning. We haven’t created a parallel accessible exhibition: it’s the same exhibition with some additional accessible features.”



The exhibition Our Universe: Back to the Big Bang features a tactile timeline of the Universe. (Image: CERN)

The permanent Science Gateway exhibitions comprise three parts: Discover CERN, Our Universe and Quantum World. Many installations incorporate an audio description and many also include tactile features, where visitors can touch and explore to interpret the content for themselves.

“One of the exhibitions that impressed me the most was Discover CERN, where I was able to go around the tunnel that the particles pass through and, thanks to the guide’s description, experience the entire journey they take. It was impressive,” says Anne Gaugaz, one of the ABA visitors with visual impairments. “I also liked, in Back to the Big Bang, being able to touch the planets of the solar system that were in relief. That helped me to understand how big they were.”

“Thanks to the guided tour, I enjoyed my visit to the Science Gateway exhibitions, in particular the accelerator exhibition and the part that takes us back in time to the Big Bang. It was very interesting and informative,” says Bertram Paul, a blind member of the ABA group who also worked with

the exhibitions team to advise on the audio descriptions.

While the visit in November was a success, there is still work to be done. “Of course, it’s an evolving process,” Jost continues. “Some more alterations need to be made, so we will keep in touch.”



A visitor from ABA touring the Discover CERN: Collide exhibition with his guide and guide dog. (Image: CERN)

Content designed for people with visual impairments is just one aspect of Science Gateway’s effort to be accessible to all. And to complement the exhibitions, there is now a new [CERN guides course](https://guides.web.cern.ch/join) focusing on accessibility, which is run by another local group, [Culture Accessible](https://guides.web.cern.ch/join). Visit this link to find out how you can become a Science Gateway guide: <https://guides.web.cern.ch/join>.

For colleagues across CERN who are interested in making their work more accessible, the exhibitions team can share more about the process. As Emma recommends, “It’s always easier to ensure accessibility when you build it in from the start of your project. Working together with the community is essential: it brings creative and sometimes unexpected solutions that often work better for everyone.”

Naomi Dinmore

The CERN Accelerator School celebrates 40 years

In 1983, when textbooks on accelerator physics were sparse, the CERN Accelerator School was born. See how its educational approach continues to make an impact



Marking the 40th anniversary of the CERN Accelerator School on the CERN lawn on 14 September 2023. (Image: CERN)

The CERN Accelerator School (CAS) celebrated its 40th anniversary in the sunshine last September and with its contributors this January.

Back in the 1980s, a group of CERN scientists and engineers saw the need for an educational training programme in the rapidly evolving field of accelerator physics and technology. Textbooks on accelerator physics were sparse at the time, and courses at universities were practically non-existent. As Herwig Schopper, then CERN Director-General, put it: “An enormous amount of expertise is stored in the brains of quite a number of people

[...]. However, very little of this knowledge has so far been documented or published in book form.” It was into this landscape that the CERN Accelerator School was born in 1983.

The success of CAS in Europe quickly caught the attention of the global accelerator community, leading to a surge in demand for its courses. To accommodate this growing interest, CAS began organising courses outside Europe, in Asia and the Americas, from 1985, in collaboration with other institutions and organisations working in accelerator physics.

Over its 40-year-long history, more than 6000 participants from across the globe have been trained.

Find out more about the history, impact and future of the CERN Accelerator School in the latest CERN Courier: <https://cerncourier.com/a/40-years-of-accelerating-knowledge/>

CERN Accelerator School

CERN software to become central hub for EU research

Over the last ten years, CERN-born Zenodo has evolved to store ever more scientific data. Today, a new project is set to ensure Zenodo’s position as a prime repository for EU research data

What is the link between particle physics, the study of biodiversity and historical linguistics? The thirst for knowledge? Yes, but also the tool researchers are using to store all their findings and make them available to their peers: Zenodo. For more than ten years, this CERN-born data repository has been evolving to store scientific data for ever more research communities and to This success story is about to take an even more ambitious turn with a new project: HORIZON-ZEN. Since its inception in June 2023, it has become the latest in a series of projects funded by the European Union to make the data collected by European research more findable, accessible, interoperable and reusable (FAIR). Since 2021,

adapt to the needs of more scientific disciplines. Notably, it was a key player in the COVID-19 response, providing a platform for researchers to efficiently share results, data sets and software to help the international scientific community respond to the pandemic. Today, it is used by more than 8000 research organisations worldwide.

making research data as FAIR as possible has become a requirement for all projects funded by the European Commission.

What does it mean to make data FAIR in practice? Today, this is still difficult for researchers to navigate, because FAIR are generic principles

rather than verifiable criteria. “With HORIZON-ZEN, we are striving to make being FAIR simpler and more streamlined for researchers, and we are working with scientific communities to tailor Zenodo for their specific domain,” explains Lars

Holm Nielsen, Section Leader in Open Science Repositories in CERN’s IT department.

Antoine le Gall

Watch: Music and physics: a spacetime voyage back to our origins, with Yo-Yo Ma and Fabiola Gianotti

Watch highlights of the world-renowned cellist in spoken and musical conversation with CERN Director-General

From creativity to curiosity and the wish to explore the unknown, physics and music are intrinsically linked. For one special evening held at CERN Science Gateway, world-renowned cellist Yo-Yo Ma and CERN physicist and Director-General, Fabiola Gianotti, discussed the commonalities and relations between physics and music and the human experiences of both.

The duo musically demonstrated concepts such as the mystery of dark matter, symmetries, complexity built upon simplicity, with Ma on the cello and Gianotti on the piano. Together, they played Franz Schubert’s Arpeggione Sonata, while a recording of Debussy’s La Cathédrale Engloutie accompanied striking images of the CERN experiments. To close the evening, Ma played a contemplative performance of Arvo Pärt’s Spiegel im Spiegel.



CERN’s Director General, Fabiola Gianotti (left) with renowned cellist Yo-Yo Ma (right). (Image: CERN)

Watch highlights of the evening by clicking on this link:

https://www.youtube.com/watch?v=B3f9dv7kP-I&ab_channel=CERN

Charm is better than beauty at going with the flow

New ALICE measurements shed light on the dynamics of charm and beauty particles in quark–gluon plasma

When two lead ions collide at the Large Hadron Collider (LHC), they produce an extremely hot and dense state of matter in which quarks and gluons are not confined inside composite particles called hadrons. This fireball -of particles – known as quark–gluon plasma and believed to have filled the Universe in the first few millionths of a second after the Big Bang– expands and cools down rapidly. The quarks and gluons then transform

back into hadrons, which fly out of the collision zone towards particle detectors.

In collisions where the two lead ions do not collide head on, the overlap region between the ions has an elliptic shape that leaves an imprint on the flow of hadrons. Measurements of such elliptic flow provide a powerful way to study quark–gluon plasma. In a [recent paper](#), the ALICE collaboration reported a new measurement of the elliptic flow

of hadrons containing heavy quarks, which are particularly powerful probes of the plasma.

Unlike the gluons and light quarks that make up the bulk of the quark–gluon plasma created in heavy-ion collisions, heavy charm and beauty quarks are produced in the initial stages of the collisions, before the plasma forms. They therefore interact with the plasma throughout its entire evolution, from its expansion and cooling to its transformation into hadrons. Multiple interactions with the plasma’s constituents bring heavy quarks into thermal equilibrium with the medium within a time that is proportional to the quark’s mass. Charm quarks are lighter than beauty quarks, so a shorter thermalisation time and a larger degree of thermalisation is expected for charm quarks than beauty quarks.

Once they thermalise with the plasma, charm quarks form D mesons and beauty quarks form B mesons, by combining with the medium’s light quarks (see figure above). Previous measurements have shown that the elliptic flow of such “prompt” D mesons, so named because they are produced right after the collisions, is almost as strong as that of the lightest hadrons, pions. Because the thermalisation time is expected to be longer for beauty quarks than charm quarks, the elliptic flow

of B mesons is predicted to be weaker than that of prompt D mesons.

In its recent analysis of non-head-on lead–lead collisions that occurred during Run 2 of the LHC, the ALICE collaboration measured the elliptic flow of B mesons, by measuring the flow of “non-prompt” D mesons that are produced in the decays of B mesons (see figure above). Key to the analysis was the adoption of a machine-learning technique to separate the products of the decay of non-prompt D mesons from those of the prompt ones, as well as to suppress the many background particle processes that mimic D meson production and decay.

The new measurement shows that the elliptic flow of the non-prompt D mesons is weaker than that of their prompt counterparts, in agreement with the expectation. The result sheds new light on the thermalisation of beauty quarks in the quark–gluon plasma, and paves the way for new ALICE measurements based on data from Run 3 of the LHC. With 40 times more collisions than the total recorded by ALICE in its previous periods of heavy-ion data taking, the new sample of lead–lead collisions taken in 2023 will allow the flow of charm and beauty particles to be studied in greater detail, shedding further light on their dynamics in the quark–gluon plasma.

CLOUD challenges current understanding of aerosol particle formation in polar and marine regions

The results imply that climate models are underestimating the formation rates of aerosol particles

Atmospheric aerosol particles exert a strong net cooling effect on the climate by making clouds brighter and more extensive, reflecting more sunlight back out to space. However, how aerosol particles form in the atmosphere remains poorly understood, especially in polar and marine regions. Globally, the main vapour driving particle formation is thought to be sulfuric acid, stabilised by ammonia. However, since ammonia is frequently lacking in polar and marine regions, models generally underpredict aerosol particles in these regions.

A new study from the CLOUD collaboration now challenges this view, by showing that iodine oxoacids are acting synergistically with sulfuric acid to greatly enhance the particle formation rates. The new findings, described in a paper published today in the journal *Science*, build on earlier CLOUD studies that showed that iodine oxoacids rapidly form particles even in the complete absence of sulfuric acid. The results imply that climate models are substantially underestimating the formation rates of aerosol particles in marine and polar regions.

“Our results show that climate models need to include iodine oxoacids along with sulfuric acid and other vapours,” says CLOUD spokesperson Jasper Kirkby. “This is particularly important in polar regions, which are highly sensitive to small changes in aerosol particles and clouds. Here, aerosol particles actually provide a warming effect by absorbing infrared radiation otherwise lost to space and re-radiating it back down to the surface.”

The CLOUD experiment is studying how aerosol particles form and grow from mixtures of vapours at atmospheric conditions in a large chamber. It differs from previous experiments both by maintaining ultra-low contaminants and by its precise control of all experimental parameters at conditions found in the real atmosphere. This includes the use of a CERN particle beam to simulate ions formed by galactic cosmic rays at any altitude in the troposphere.

The new CLOUD results show that iodine oxoacids greatly boost the formation rate of sulfuric acid particles. At iodine oxoacid concentrations that are typical of marine and polar regions – between 0.1 and 5 relative to those of sulfuric acid – the

CLOUD measurements show that the formation rate of sulfuric acid particles is increased by 10- to 10 000-fold compared with previous estimates.

The CLOUD team found that this increase is due to two effects: first, iodous acid substitutes for ammonia to stabilise newly formed sulfuric acid particles against evaporation and, second, iodic acid facilitates the formation of charged sulfuric acid clusters. Using quantum chemistry, the collaboration has confirmed the synergy between iodine oxoacids and sulfuric acid, and calculated particle formation rates that closely agree with the CLOUD measurements.

“Global marine iodine emissions have tripled in the past 70 years due to thinning sea ice and rising ozone concentrations, and this trend is likely to continue,” says Kirkby. “The resultant increase of marine aerosol particles and clouds, suggested by our findings, will have created a positive feedback that accelerates the loss of sea ice in polar regions, while simultaneously introducing a cooling effect at lower latitudes. The next generation of climate models will need to take iodine vapours into account.”

Computer Security: 300 computer security articles and counting

We begin the year by celebrating the 300th Bulletin article focusing on various computer-security-related topics. Three hundred hopefully informative articles about the cybersecurity situation at CERN. About best practices, guidelines and useful tools. About risks, threat scenarios and attack vectors. About new or established means of mitigation. About the workings of the Computer Security team, the services and tools it's providing and the complexity of its detection infrastructure. About policies and dos and don'ts. Three hundred articles trying to raise your awareness and help you improve your approach to computer security – the security of your laptops, smartphones and tablets, of your accounts and passwords, of your email inbox and web browser, of software programming and system development – both at CERN and at home.

While some articles were published a long time ago – the first ones were released in 2008 – they've never lost their relevance. Sometimes it's useful to delve into the past and dig out information from them; often these articles also provide guidelines for us when advising users. So for this 300th anniversary, we have updated our compilation of all the articles published so far. This compilation covers a plethora of topics, sorted into the notorious themes of “computer security”, i.e. the literal cybersecurity of computers, “mobile and cloud security”, “network and data centre security”, “account and password security”, “control systems and IoT (Internet of Things)”, “secure software development”, “data protection and privacy”, “copyrights”, “rules and policies” and more. Giving a deeper insight into the computer security landscape, these articles complement our Monthly Reports, which usually

depict the operational side of what's currently happening at CERN.

You can download this compilation here: <https://cern.ch/gbeu2>. It's public – published under CC-BY-NC-SA. So please feel free to share it with your colleagues, family and friends in order to spread the word, raise awareness and help them improve the security and protection of their own digital assets and resources!

Do you want to learn more about computer security incidents and issues at CERN? Follow our [Monthly Report](#). For further information, questions or help, check our website or contact us at Computer.Security@cern.ch.

Computer Security Team

Official news

Update on CERN Safety Rules

The CERN Safety Rules listed below have been published on the CERN website dedicated to the Safety Rules :

- Safety Regulation SR-SIM “Responsibilities in Matters of Safety Incident Management at CERN” - version 2 EDMS 2583792. The version 2 of the SR-SIM cancels and replaces version 1 of the same document. This new version takes into account lessons learnt from the last two years of application of the SR-SIM version 1. It introduces the possibility of a more formal and in-depth analysis of a serious Safety Incident and a specific notification and analysis procedure for Safety Incidents occurring on category 1 worksites.
- Safety Regulation SR-NIR “Protection of Persons from Exposure to Non-ionising

Radiation” - EDMS 2974850. The SR-NIR covers a large variety of hazards, notably static magnetic field and lasers. It defines the general prevention and protection principles applicable to all of them.

- General Safety Instruction GSI-NIR-1 “Protection of Persons from Exposure to Static Magnetic Fields” - EDMS 2974732 This GSI cancels and replaces IS 36 “Safety Rules for the Use of Static Magnetic Fields at CERN”.

The CERN Safety Rules apply to all persons under the Director General's authority. They are available under the following link: <http://www.cern.ch/safety-rules>.

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

1 Feb 12:00 | At CERN | Online | Medical Service | EN and FR

[Cancer awareness campaign](#) (more [information](#))

2 Feb 16:30 | Knowledge Sharing | Main Auditorium | CERN Colloquium | EN

[US Particle Physics for the Next Ten Years](#)

5-8 Feb | Knowledge Sharing | Council Chamber | Academic Training Lectures | EN

[Science for Nuclear Arms Control](#)

8 Feb 10:30 | At CERN | Main Auditorium | Human Resources | EN and FR

[CERN Staff and Fellow survey recommendations](#)
[| Recommandations de l'enquête auprès des titulaires et boursiers du CERN](#)

9-11 Feb | Knowledge Sharing | CERN Science Gateway | Alumni Relations

[CERN Alumni 3rd collisions](#)

27 Feb 11:00 | Physics | Main Auditorium | EP Seminar | EN

[IceCube](#)

1 Mar 14:00 | Knowledge Sharing | Main Auditorium | Symposium | EN

[Herwig Schopper – a century in physics](#)

7 Mar 20:00 | Knowledge Sharing | CERN Science Gateway | CERN70 | EN

[CERN70 public event: From particle physics to medicine](#) (more [CERN70](#) events)

8 Mar 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN

[L&D Micro-talk 9 – Intergenerational dynamics in the workplace](#)

Registration now open

11-14 Feb | Accelerators | Wiener Neustadt (AT) | MedAustron, GSI, I.FAST | EN

[5th Slow Extraction Workshop](#)

11-19 Mar | Computing | CERN | IT Department | EN

[TechWeekStorage24](#)

2-15 Jun | Accelerators | Sint-Michielsgestel (NL) | CERN Accelerator School | EN

[Mechanical & Materials Engineering for Particle Accelerators and Detectors](#)

12-25 Jun | Physics | Nakhon Pathom (TH) | Asia-Europe-Pacific School of High-Energy Physics (AEPSHEP) | EN

[2024 Asia-Europe-Pacific School of High-Energy Physics](#)

23 Jul-1 Aug | Knowledge Sharing | European Scientific Institute (FR) | I.FAST | EN

[Accelerators for healthcare? 10-day innovation challenge](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events

happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch.

Preventing falls and other seasonal accidents now that winter is upon us

In winter, the snow and ice can turn pavements and road surfaces into skating rinks, increasing the risk of slips and injuries. At this time of year, the CERN infirmary sees more people coming in with problems after a fall. During these winter months, it's important to wrap up well (wear warm shoes with a good grip, a hat, gloves, etc.), use the

appropriate means of transport (public transport or a car fitted with winter tyres rather than a bike or motorbike) and take great care when travelling. Winter sports are also a common source of accidents. Over 33 000 employees in Switzerland are injured in such accidents every year (Suva, the Swiss national accident insurance fund, 2023).

According to a report published by the French Observatoire d'accidentologie des sports d'hiver, more than 122 000 people were injured while practising winter sports in France during the 2021–2022 season.

In the case of downhill skiing, as shown in the image below, which was published by Suva in 2023, the hips, legs, knees and feet are the most commonly injured body parts (45%), followed by the shoulders, arms, elbows, hands and fingers at 33%. The latter figure rises to 42% in the case of snowboarding.

Whether or not you're a fan of winter sports, and whichever sport you choose, make sure that you're prepared and have the right kit. Here are some basic tips for skiing and snowboarding in particular:

- Good physical preparation is key to reducing the risk of an accident. Strengthen your muscles and do cardiovascular workouts (see advice (in French) from Suva on making the most of the season).
- Get your ski equipment checked every year; a lot of sprained knees are caused by badly adjusted fastenings or boots.

- Wearing a helmet reduces the risk of head injury in the event of an accident by 35%: get a helmet that meets the standards and is the right size for your head.
- Warm up / stretch before hitting the slopes.
- Wear wrist guards if you're a snowboarder.
- Eat regularly and drink hot drinks to stay hydrated.
- Slow down.
- Have a great winter!

For more information, check out the following links (in French):

- [Top 3 des accidents de sports d'hiver et comment les soigner \(medisafe.fr\)](https://www.medisafe.fr/actualites/top-3-des-accidents-de-sports-d-hiver-et-comment-les-soigner)
- [Sports d'hiver | Pratiquer ski et snowboard sans accident \(suva.ch\)](https://www.suva.ch/fr/sports-d-hiver/pratiquer-ski-et-snowboard-sans-accident)
- [Tout schuss sur les pistes : les blessures liées à la pratique du ski - Chirurgie Orthopédique \(chirurgie-orthopedique.fr\)](https://www.chirurgie-orthopedique.fr/tout-schuss-sur-les-pistes-les-blessures-liees-a-la-pratique-du-ski)

CERN Medical Service

Ombud's corner

The potential harm of anonymous complaints

I very rarely receive anonymous complaints or accusations of inappropriate behaviour or misconduct, but it does happen. These complaints always arrive by emails that can't be traced back to the sender, should anyone want to do so.

Because they feel protected from any kind of consequence, the alleged victims or witnesses often vent the strongest feelings of anger, frustration, indignation, envy or animosity. And since the senders avoid giving details that could identify them, what they write is often quite vague.

All sorts of things may hide behind an anonymous complaint, including:

- a colleague who is in good faith and truly affected by a breach of the Code of Conduct, such as harassment, and particularly concerned about possible retaliation, especially if the issue is with a supervisor or someone high up in the hierarchy;
- someone wanting to harm another colleague by triggering an investigation against them;

- several anonymous complaints targeting the same person in a short time, in which case the possibility of a concerted attempt to impact that person's career, for whatever reason, must be considered; or
- simply someone enjoying the art of accusing and spreading rumours.

The impact of an anonymous complaint on the target is significantly stronger than when the person knows who has complained and why. With anonymous complaints, it is impossible to react, defend oneself or, in some cases, to understand why a management or disciplinary process has been initiated. This is extremely destabilising and may also put the person accused at risk.

No one can respond to accusations when they don't know who is accusing them and when they are given only vague information about what allegedly happened and when.

Anonymous complaints also deprive the Ombud of the opportunity to discuss with the complainant their fear of retaliation and to understand how and why such a fear is grounded in their work environment. They also deprive the complainant of having their case reflected in the trends that are reported in an entirely anonymous way to the top management as part of the Ombud's annual report.

When I received anonymous complaints, I once again had to carefully reflect on the Ombud's role. And after having deeply considered how to address them, I have come to the conclusion that

I can only acknowledge them but decline to act on them in any way.

I respond to the anonymous senders with a copy of the Ombud's mandate to remind them that my role is centred on informal dispute resolution and explain why I will not act on their email. I also remind them that the CERN Ombud is the closest they have at their disposal to a whistleblowing channel, as the CERN Ombud will not – under any circumstances – reveal their identity or do anything without their authorisation. I also invite them to a meeting in my office so that they can describe the case. If they do not want to risk being seen in my office, I suggest a Zoom meeting.

The CERN Ombud will not act on anonymous complaints. If your complaints or accusations are in good faith and if you are concerned about possible retaliation, I encourage you to contact the Ombud at ombud@cern.ch. This will give you the opportunity to expose the breach of the Code of Conduct that you are aware of, while fully preserving your anonymity.

Laure Esteveny

I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost team members at <https://mattermost.web.cern.ch/cern-ombud/>.

More information on the role of the CERN Ombud and how to contact her can be found at <https://ombud.web.cern.ch>